

PHASE-ADAPTIVE QUANTIZATION

Design space exploration of 4-bit MMA rescale pipelines for AI accelerators

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Hardware-aware quantization for LLM, VLM, and VLA inference workloads

Thesis: one bundled 4-bit format is not globally optimal. A known accelerator target should choose the rescale datapath by workload and inference phase.

60+60

completed hardware + accuracy rows

50-60%

weighted energy cut vs NVFP4-like C7

3

selected datapaths across six cells

1. MOTIVATION

LLM inference alternates between prefill and decode, but common 4-bit formats bundle a fixed rescale pipeline for both.

Decode, M=1

memory-bandwidth-bound; weight movement dominates, so compression and metadata overhead are central.

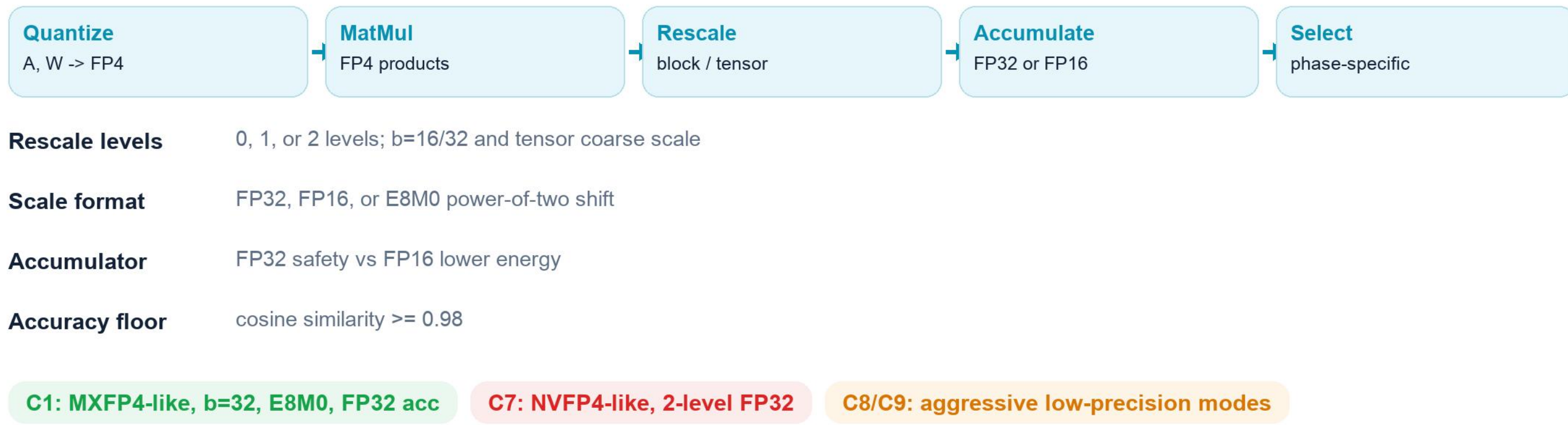
Prefill, M=128

compute/rescale-heavy; FP32 rescale and accumulation can dominate MMA energy.

Design gap

NVFP4/MXFP4 are useful reference points, but each fixes levels, block size, scale format, and accumulator precision.

2. UNDERLYING PRINCIPLES



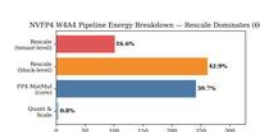
3. EVALUATION

Workload	Shape (M,N,K)	Role
LLM FFN	decode (1,11008,4096) / prefill (128,11008,4096)	Llama-style feed-forward GEMM
VLM Vision	decode (1,3072,3072) / prefill (128,3072,3072)	balanced vision GEMM
VLA Action	decode (1,256,4096) / prefill (128,256,4096)	latency-sensitive action head

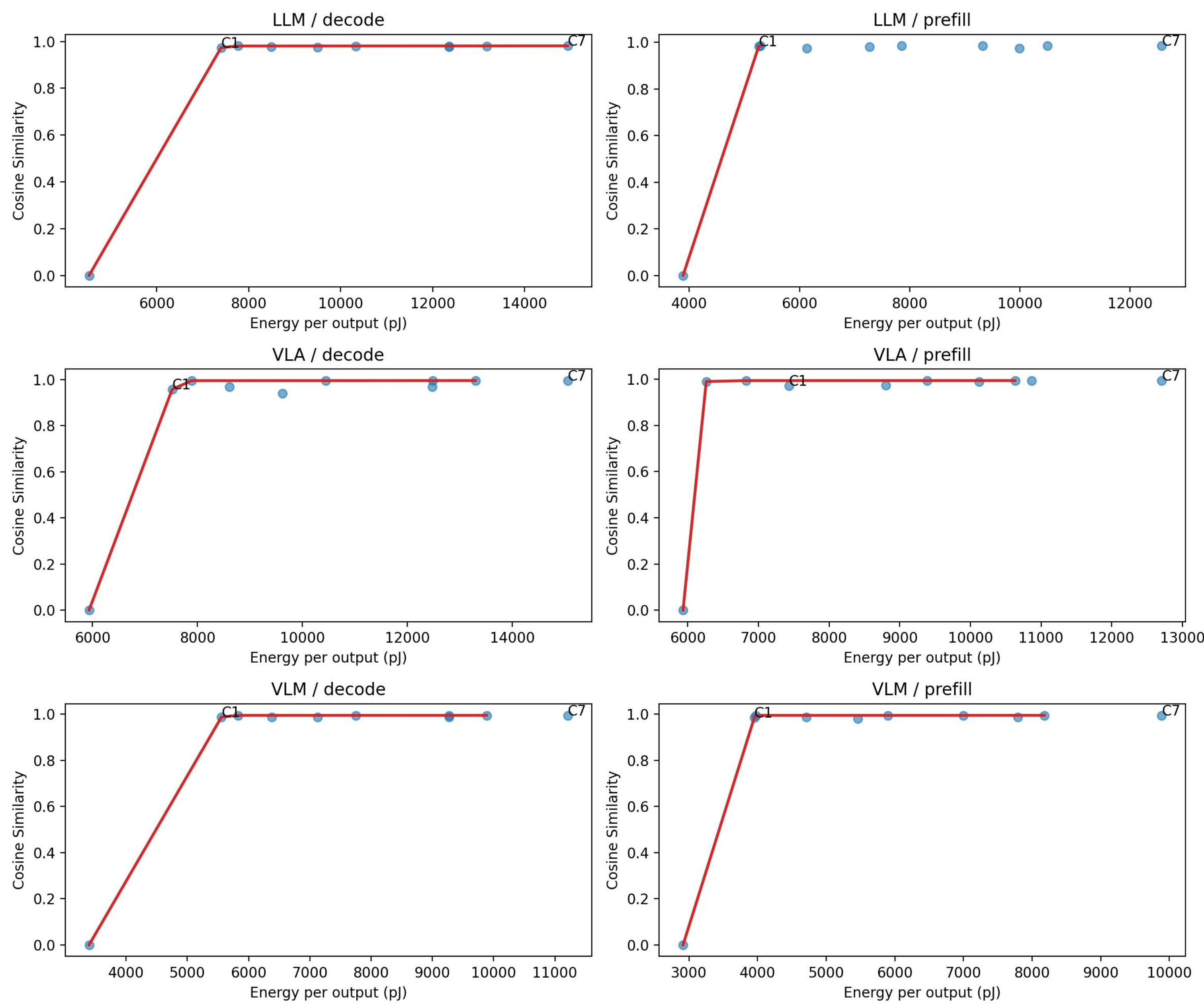
Hardware AccelForge/Timeloop-style cost modeling extracts total energy, energy per output, latency, area, and bottleneck component.

Accuracy A deterministic quantized-MMA emulator compares layer outputs against FP16 using cosine similarity and SQNR.

Sweep 10 configs x 3 workloads x 2 phases = 60 hardware rows + 60 accuracy rows; all completed OK.



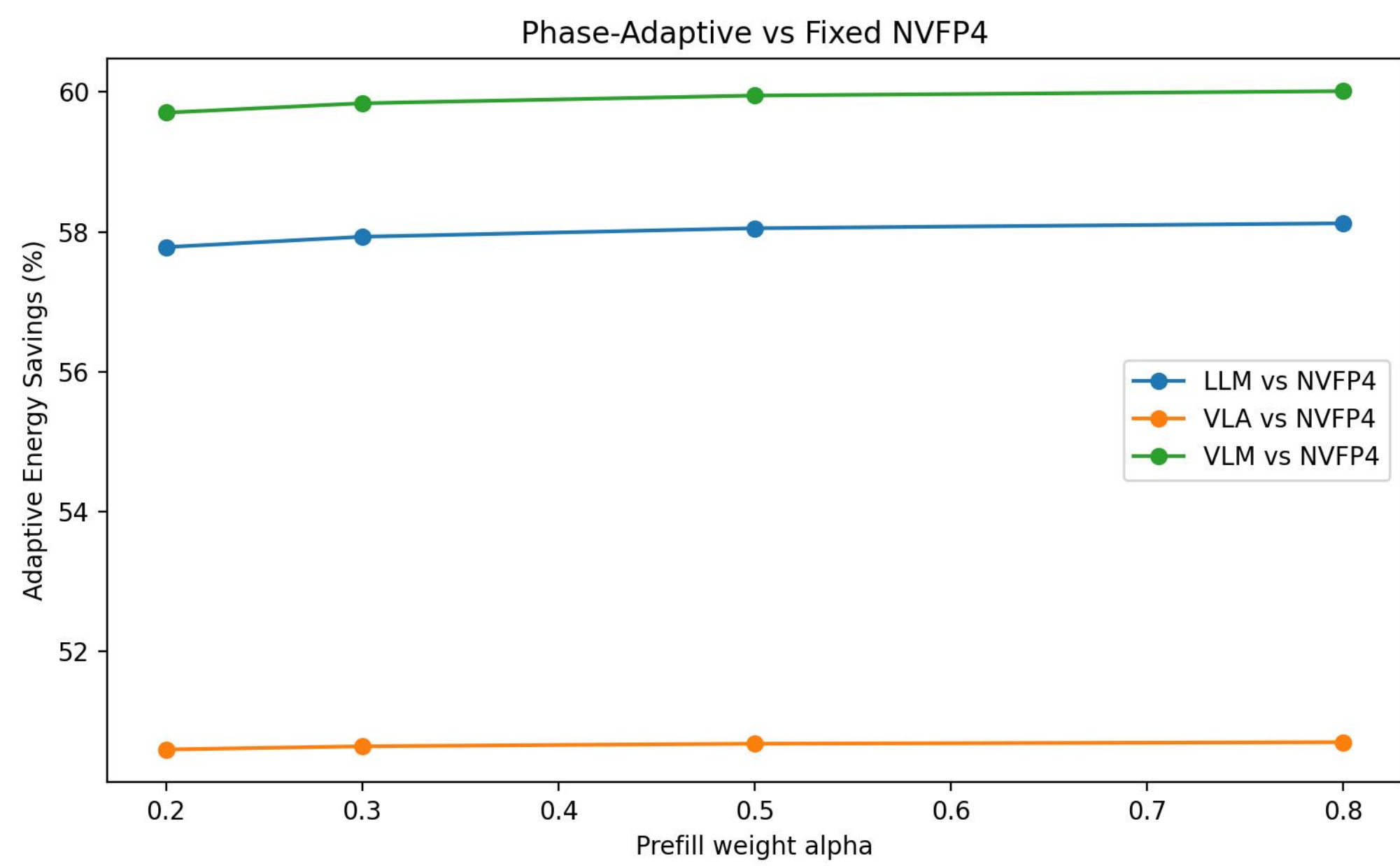
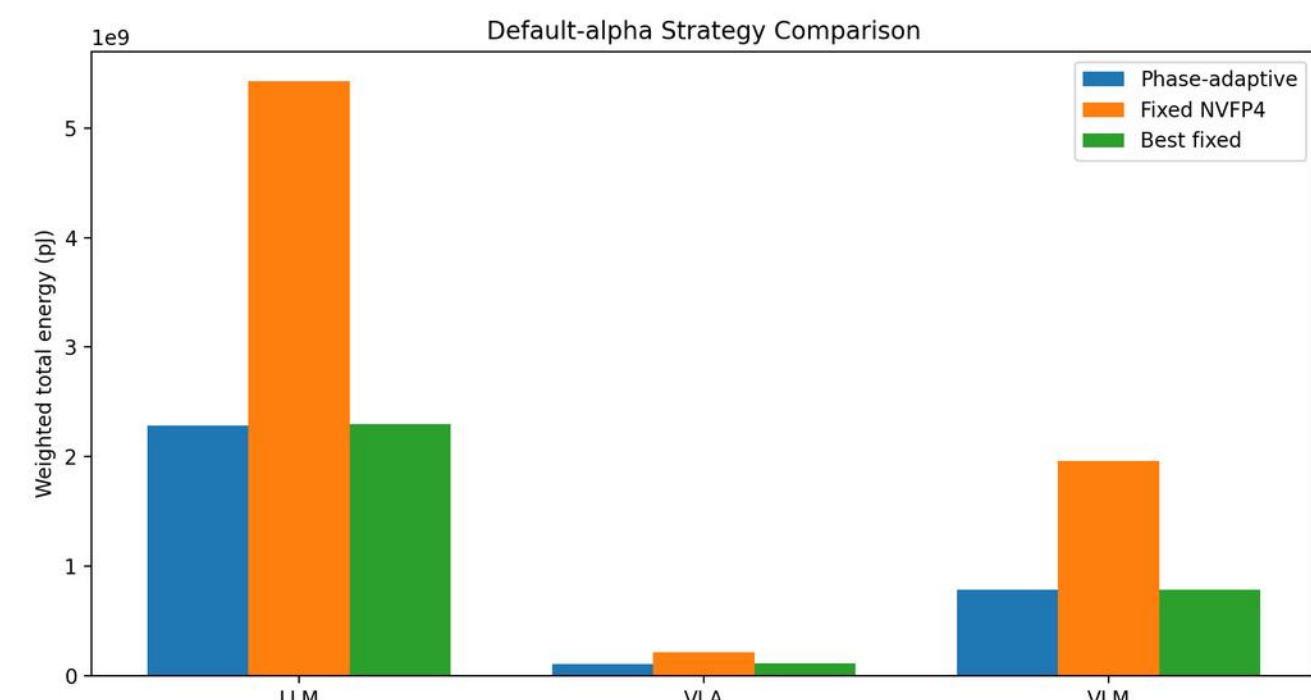
4. PARETO RESULTS



Each panel plots energy per output against cosine similarity. C7 is the NVFP4-like reference; C1 is the MXFP4-like reference.

6. RESULTS ANALYSIS

LLM 58% less weighted energy vs C7	VLM 60% less weighted energy vs C7	VLA 51% less weighted energy vs C7
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Compared with best-fixed, adaptation is selective: VLA gains about 8.2%, LLM about 0.6%, and VLM 0% because both VLM phases already select C1.

5. BEST CONFIGS @ COSINE ≥ 0.98

Workload	Decode	Prefill	Analysis
LLM	C8 7,767 pJ/out cos 0.981	C1 5,262 pJ/out cos 0.984	phase-adaptive: C8 clears decode accuracy; C1 is the cheaper prefill knee.
VLM	C1 5,557 pJ/out cos 0.987	C1 3,952 pJ/out cos 0.984	single simple mode: C1 wins both phases because the balanced GEMM tolerates E8M0 scaling.
VLA	C8 7,891 pJ/out cos 0.995	C9 6,263 pJ/out cos 0.990	phase-adaptive: C8 fits decode, while C9's hybrid scale path wins prefill.

Result: no single configuration wins all six workload-phase cells; phase adaptation matters when decode and prefill stress different hardware costs.

7. REFERENCES

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- [2] NVIDIA, Blackwell Architecture and NVFP4 Tensor Core documentation.
- [3] Parashar et al., Timeloop: A Systematic Approach to DNN Accelerator Evaluation, ISPASS 2019.
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